

Exercise 6.2

Mean, Median, Mode & Range

1. The following sets of numbers are arranged in ascending order.

Find the median for each set.

- a) 2, 2, 2, 3, 7, 7, 11
- b) 10, 10, 15, 16, 19, 20, 21, 21, 23
- c) 7.5, 7.6, 7.6, 7.7, 7.7, 7.8, 7.8, 7.9, 8.0, 8.1, 8.1, 8.5, 8.5
- d) 2, 3, 4, 5, 7, 7, 10, 10
- e) 12, 15, 15, 16, 19, 19, 20, 24
- f) 5, 6, 6, 7, 8, 9, 9, 9, 10, 10, 10, 11

2. Locate the median of each of the sets of data below.

Arrange them in ascending order first.

- a) 5, 6, 8, 8, 9, 5, 4
- b) 72, 85, 87, 80, 90, 92, 85, 89, 89
- c) 3, 1, 5, 1, 5, 6, 4, 5
- d) 17, 24, 20, 17, 18, 17, 18, 19

3. Consider three distributions:

A: 8, 6, 8, 8, 8, 10, 8, 8, 8

B: 8, 12, 4, 5, 8, 11, 8

C: 7, 9, 3, 9, 13, 2, 14, 11, 7, 7, 9, 5, 11, 5

- a) Find the mean of each distribution.

Only two of these distributions have a mode.

- b) Find these modes.

Also,

- c) calculate the range for each distribution and determine the distribution with the greatest spread and the smallest spread.

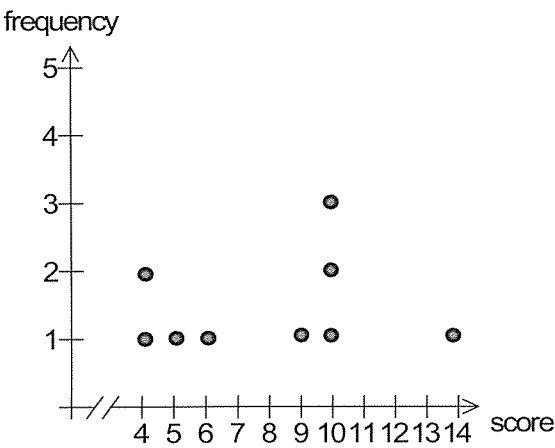
4. Find the mean, median, mode and range for each set of scores.

Round your answers to one decimal place where necessary.

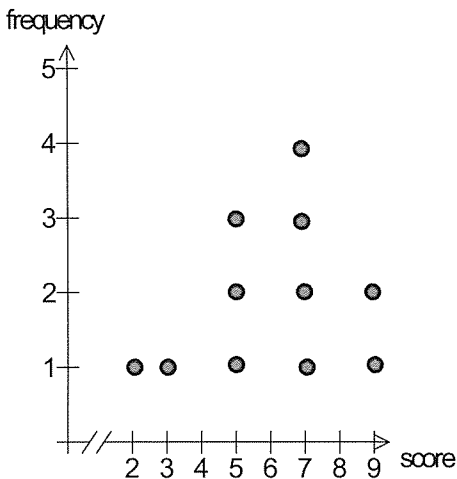
- a) 15, 20, 20, 65, 80, 142
- b) 102, 95, 95, 76, 108, 120
- c) 7, 7, 10, 10, 10, 18, 400

5. Find the mean, median, mode and range of the scores shown in the dot plots below.

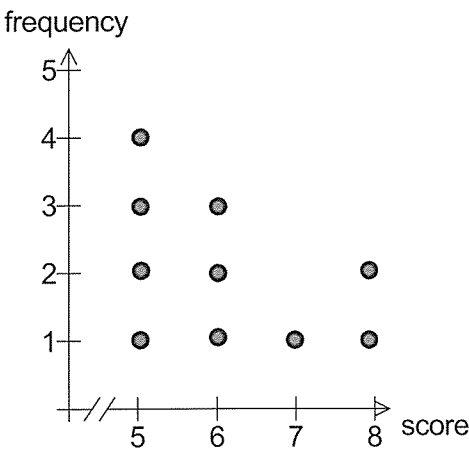
a)



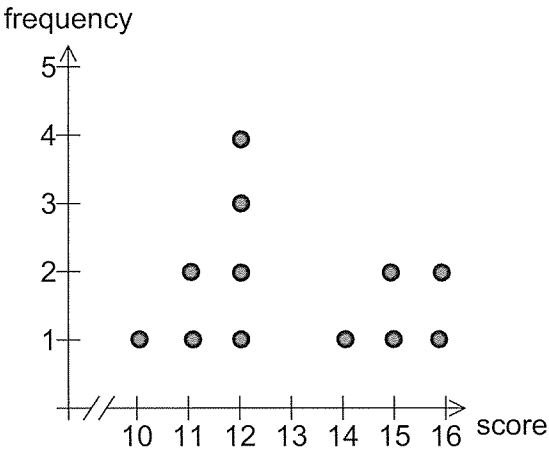
b)



c)



d)



Also,
e) which distribution has the greatest range?